

From Terrestrial Laser Scans to Queryable Robot Knowledge: A VLM-Verified Framework for Incremental 3D Semantic Modeling

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Abstract

Indoor service and inspection robots need environment knowledge that is queryable, reliable, and maintainable as the site changes, yet open-vocabulary 3D pipelines propagate unverified labels downstream and build-once scene models offer no maintenance path. We present TLS-SMF, a TLS-driven semantic modeling framework that converts registered terrestrial laser scans into a confidence-aware semantic knowledge graph and keeps it current across repeated scans. TLS-SMF combines a deterministic geometric backbone whose re-runs are byte-identical on the same host; multi-view vision-language-model verification with automatic escalation, which retains unresolved objects as explicitly unverified; and an identity-preserving incremental update that applies confidence-gated node-level upserts to mint-once node identities without rebuilding the graph. On two industrial indoor scenes, escalation lifts verification accuracy from 75.0% to 86.1% and from 78.1% to 84.4%, with 94.4–100% precision for unanimous votes; on the showroom scene, structurally gated querying answers 83.3% of benchmark questions versus 52.1% for an ungrounded LLM while blocking unverified-label hallucination traps; and selective re-verification prices a changed re-scan at 3.1% of a full rebuild (0% when unchanged). A real three-epoch study with a controlled-change re-scan detects 2/2 physical changes at geometry level while preserving object identity across all three epochs. Beyond one-shot verification, automatic re-split gates (diffuseness and wall-compound triggers), a structure-condition detection ensemble, and an owner-in-the-loop revision path keep the graph corrigible in use, with the management console, the mission mediator, and a simulation twin all derived from the one revised store. A full re-run with no owner input recovers 85% of the resulting owner-audited 46-object inventory, versus 28% for the verified path alone; the residual type errors concentrate where verifier bias is correlated across conditions, which is precisely the role owner evidence is designed to fill.

Keywords: terrestrial laser scanning; semantic mapping; knowledge graph; vision-language models; incremental update; mobile robots

1 Introduction

Mobile robots deployed in industrial interiors increasingly need more than a map: they need environment *knowledge*—which objects exist, where they are, what they afford—that can be queried in task language, trusted enough to act on, and kept current as the site changes. Terrestrial laser scanning (TLS) is an attractive source for such knowledge: survey-grade scanners deliver millimeter-accurate, globally registered geometry of an entire site in minutes, and are already routine in construction and facility management. Yet the path from a raw scan to knowledge a robot can safely use is broken in three places.

First, *semantic labels do not survive contact with industrial scenes*. Open-vocabulary 3D classifiers trained on web-scale or indoor-furniture distributions degrade sharply on industrial content: in our experiments a state-of-the-art point-text embedding model misclassifies most equipment, and—counterintuitively—restricting its vocabulary to the classes actually present makes accuracy *worse* (66.7% → 33.3%, Sec. 5.6), indicating an embedding-level domain gap rather than a class-list problem. Pipelines that propagate such labels downstream hand the robot confident fictions.

Second, *ungrounded language interfaces hallucinate*. Letting an LLM answer questions about a scene from its own priors, or from an unvalidated label dump, produces fluent but unsupported claims—20.8%

of atomic factual claims in our LLM-only benchmark condition (Sec. 5.4). For a robot, a hallucinated fire extinguisher is not a harmless chat artifact.

Third, *scene models are built once and abandoned*. Most semantic mapping pipelines produce a single-epoch artifact with no principled update path: a re-scan either rebuilds everything—losing object identities that downstream planners and logs refer to—or is not integrated at all. Real sites drift continuously; in our own test room, roughly one third of object geometry changed within two weeks (Sec. 5.5).

This paper presents **TLS-SMF** (TLS-driven Semantic Modeling Framework), which addresses all three failures with one design principle: *separate what can be made deterministic from what must remain probabilistic, and control the probabilistic part with explicit confidence machinery*. Concretely, we contribute:

1. **A deterministic geometric and explicit-modeling backbone.** From registered TLS input, seeded-RANSAC structure removal, fixed-parameter clustering with a conservative giant-split and a structure-remnant filter, and PCA-based explicit modeling run fully automatically; same-host re-runs are byte-identical. Determinism is not an implementation nicety: it is what makes verification results carryable, diffs meaningful, and update costs proportional to actual change.
2. **Confidence-aware multi-view VLM verification with automatic escalation.** Each object is classified per-view under a forced answer schema; votes are fused confidence-weighted; split votes escalate automatically to a wider view set; and objects that still fail to converge are retained as explicitly **unverified**—never silently dropped and never resolved by a human in the loop.
3. **Identity-preserving incremental knowledge-graph update.** Node IDs are minted once and carried by two-pass matching (spatial, then type-constrained move detection); re-scans apply node-level **unchanged/updated/moved/inserted/absent** upserts with per-revision history, and unchanged objects re-verify for free by content-hash carry-over.
4. **Corrective decomposition and condition-varied detection.** Diffuseness and wall-compound gates turn “too messy to be an object” from a discard decision into an automatic re-splitting trigger (wall-band stripping, support-plane and panel-bank cuts), and a structure-condition ensemble re-detects under varied wall/ceiling removal, recovering wall-flush objects any single condition amputates while exposing label instability as a per-object attribute.
5. **Owner-in-the-loop revisions over a single semantic store.** The room occupant’s refutations, restorations, and corrections of labels, geometry, and attributes enter as highest-trust but time-variant evidence through the same revision mechanism as re-scans; the management console, the mission mediator, and an Isaac Sim twin are all derived from the one revisioned graph, so a correction propagates to every consumer from a single write.

We validate the framework on two industrial indoor scenes with owner-confirmed ground truth, through five benchmark families: segmentation repeatability and sensitivity, verification accuracy and escalation cost, a four-condition knowledge-grounded query benchmark with hallucination traps, synthetic update scenarios, and a real three-epoch study of one room spanning a manual scan, an autonomous exploration scan, and a controlled-change re-scan at 99.6% coverage. Beyond the headline gains—escalation +11.1/+6.3 points, gated querying +31 points over LLM-only with traps blocked, re-verification at 3.1% of rebuild cost—the study surfaces failure modes we believe are as valuable as the successes: phantom objects from ceiling soffits, unanimous-but-wrong verifications, clutter absorption of small low-reflectance objects, and a coverage paradox in which a *better* scan over-merges density-based clustering. Each is analyzed and either fixed by a method component or reported as a bounded limitation.

The framework extends our ICTC 2026 multi-view VLM classification study with escalation, structural gating, identity matching, incremental update, and the multi-epoch evaluation; the knowledge-graph lineage follows our ISIS 2025 work. Both are cited and the deltas are stated explicitly in Sec. 2.

2 Related Work

2.1 3D Semantic Mapping and Scene Graphs

3D scene graphs organize metric reconstructions into object-level nodes with semantic attributes and inter-object relations [1], and recent open-vocabulary variants attach language-queryable features to nodes and edges [2, 3], enabling LLM-driven mobile manipulation over the graph [4]. These systems target flexible *construction*; verification of what enters the graph and maintenance after construction receive less attention. Within the TOSM (triplet ontological semantic model) lineage that this work extends, the semantic modeling framework and its successor DK-SMF [5] automate object *and place* modeling

Table 1: Positioning against DK-SMF [5].

Axis	DK-SMF	This work
Sensing	RGB-D exploration	registered TLS (BLK360)
Label trust	domain-knowledge filtering	multi-view verification + escalation
Lifecycle	build-once	mint-once IDs, incremental upsert
Place semantics	SLIC + LLM over detections	SLIC + ring codes over verified objects

from RGB-D exploration, combining zero-shot detection and VQA with domain-knowledge filtering and LLM-derived place semantics over SLIC-segmented regions. Our framework—TLS-SMF, named within the same lineage’s convention of prefixing the differentiating axis—adopts the same target representation (TOSM object/place layers, VDA5050-style export) but differs on four axes, summarized in Table 1: a survey-grade TLS backbone instead of RGB-D streams; multi-view VLM *verification with escalation* instead of domain-knowledge hallucination suppression; identity-preserving *incremental* upserts instead of build-once modeling; and a place layer derived from *verified objects only*, so the confidence gate propagates across layers.

2.2 Open-Vocabulary 3D Recognition and Its Limits

Point–text embedding models such as Uni3D [6] and PointCLIP V2 [7] transfer web-scale vision–language supervision to point clouds and perform well on curated indoor benchmarks. On industrial TLS content, however, we observe the transfer breaking down: in a 30-cluster pilot, PointCLIP V2 collapsed to a single class for 23 of 30 clusters (0% top-1), and Uni3D’s accuracy *drops* as the vocabulary is narrowed toward the true class list (66.7% $\rightarrow$ 33.3%, Sec. 5.7)—the embedding cannot separate the domain’s classes regardless of prompt curation. We therefore treat open-vocabulary output as a provisional prior to be verified, not a label source.

2.3 Instance Segmentation: Learned versus Geometric

Learned 3D instance segmentation (SPFormer [8], PointTransformer V3 [9]) and promptable point segmentation (Point-SAM [10]) dominate benchmark suites but inherit their training distributions; our transfer attempts to industrial TLS content failed without target-domain fine-tuning, for which no labels exist in our setting. We instead use a deliberately simple geometric decomposition—seeded plane removal plus fixed-parameter density clustering—and position it not as a novel algorithm but as a *reproducible decomposition* whose determinism underwrites verification carry-over and incremental diffing (Sec. 4.2). Point-SAM appears in our pipeline only as a conditionally adopted refinement of flagged merged clusters, under a pre-registered adoption rule (Sec. 5.2).

2.4 Multi-View VLM Verification

Using VLMs to label rendered 3D content is increasingly common; our prior work [11] established multi-view forced-schema VLM classification of TLS object clusters with confidence-weighted late fusion. This paper extends that mechanism into a verification system: automatic escalation of split votes to a wider view set, explicit retention of unresolved objects as **unverified**, tier-precision analysis (including unanimous-but-wrong failure cases), and integration of the resulting status into gating, update, and place layers. The comparison in Sec. 5.3 quantifies the delta over the 4-view baseline.

2.5 Knowledge-Grounded Querying and Hallucination Mitigation

Grounding LLM answers in retrieved structure reduces but does not eliminate unsupported claims [12], and instruction-based guardrails are known to be brittle. Our knowledge-graph lineage [13] observed hallucinated environment facts when querying ungrounded LLMs about mapped spaces. We contribute a structural mitigation: rather than instructing the model to respect verification status, the query-time view demotes unverified types to **unknown** and quarantines the failed label in a schema-documented field—an intervention in the data contract whose effect we isolate in an ablation (traps 1/4 $\rightarrow$ 4/4, Sec. 5.7).

Table 2: Capability comparison of environment representations.

	2D map	3D scan	3D semantic KG (this work)
Metric navigation	✓	✓	✓ (derived)
Object geometry	–	✓	✓ (explicit models)
Task-language query	–	–	✓ (gated)
Per-fact confidence	–	–	✓ (verification tiers)
Change maintenance	re-map	re-scan	node-level upsert
Place semantics	–	–	✓ (verified ring codes)

2.6 Incremental Semantic-Map Maintenance

Change detection in point clouds and lifelong mapping address geometric currency [14], and object-permanence reasoning tracks instances through rearrangement in embodied settings [15]. What is rare is maintenance of a *semantic knowledge graph* whose nodes carry verified labels and stable identities across scan epochs: scan-once-build-once remains the default in TLS-derived modeling. Our mint-once identity policy with two-pass matching and confidence-gated node-level upserts (Secs. 4.5–4.6) targets this gap, and Sec. 5.5 evaluates it on synthetic scenarios and a real three-epoch sequence with a controlled-change protocol.

2.7 Map Representations for Robot Tasking

Table 2 contrasts the representations a deployment can choose from. A 2D occupancy map supports navigation but neither queryability nor semantics; a raw 3D scan adds geometry but no task language; a verified semantic knowledge graph adds queryability, per-fact confidence, and—with this work—a maintenance path. The table is analytical; its query and update columns are substantiated by Secs. 5.4 and 5.5.

3 System Overview

Figure 1 shows the framework as a staged pipeline with four loops. A registered TLS scan enters the *deterministic geometric backbone* (blue in the figure): preprocessing with room-scoped structure removal, object decomposition with remnant filtering, automatic *re-split gates* that turn diffuse webs and wall-flush compounds into decomposition triggers rather than discards, and explicit modeling. A parallel *structure-condition ensemble* re-detects under varied removal conditions, contributing wall-flush candidates that any single condition amputates and a per-object label-stability signal. The backbone’s output feeds the *probabilistic semantic stages* (orange): multi-view VLM verification with a height-level physical prior and an internal escalation loop, then confidence-gated integration into the knowledge graph. The graph is the *single semantic store*: the management console, the mission mediator, and the Isaac Sim twin are all derived from the same revisioned file. Two further loops close over it: the *update loop* (a later scan re-enters the backbone, is matched against existing node identities, and applies a node-level upsert rather than a rebuild) and the *owner loop* (purple), in which the room occupant refutes, restores, and corrects labels, geometry, and attributes as highest-trust—but time-variant—evidence, each intervention landing as a provenance-carrying revision.

The color separation in the architecture is a claim boundary, not a drawing convention. Everything in the backbone is seeded and parameter-fixed; re-running it on the same host reproduces every retained cluster byte-identically (measured in Sec. 5.2; across hosts we define reproducibility numerically, within floating-point tolerance). Everything downstream of the first VLM call is stochastic; there we claim *control*—schema-forced outputs, multi-view voting, automatic escalation, and confidence gating—rather than reproducibility.

Each object node carries the three TOSM description layers: a *symbolic* model (type, name, status), an *explicit* model (pose, extents, yaw, color), and an *implicit* model (movability, openability, key-object status). Verified and unverified content flow through the same graph but are separated at every interface: verification status is part of the node schema, gating is applied structurally at query time (Sec. 4.7), and the place layer is derived from verified objects only (Sec. 4.8).

The framework is fully automated *after registered TLS input*: scanner operation and vendor multi-setup registration are manual; everything from E57 to queryable graph—including escalation decisions and

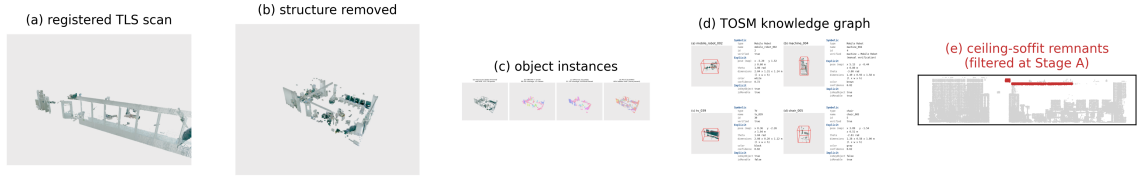


Figure 2: Geometric backbone: preprocessing, plane removal, decomposition, and the structure-remnant filter (inset: ceiling-soffit phantom before/after).

4.2 Geometric Object Decomposition and Structure-Remnant Filtering

The structure-free cloud is partitioned into object candidates by fixed-parameter DBSCAN ($\epsilon = 0.30$ m, 100-point minimum), followed by a *conservative giant-split*: any cluster whose horizontal footprint exceeds 2.5 m is re-clustered at a tighter $\epsilon = 0.20$ m, and the split is accepted only if it yields at least two sub-clusters of 100+ points—separating merged neighbors without shattering genuinely large equipment. We position this stage deliberately: it is not a novel segmentation algorithm but a *reproducible decomposition designed for verification and incremental diffing*. Its value is that identical inputs yield identical clusters with stable ordering, which later stages exploit for result carry-over and change detection (Sec. 4.5). The parameter sensitivity of this choice is quantified in Sec. 5.2.

Structure-remnant filter. Real buildings violate the implicit assumption that structure is planar and objects are not. Our showroom scene has a three-level ceiling ($z = 2.48/2.18/1.95$ m): RANSAC removes each horizontal ceiling plane, but the *vertical soffit bands* at the level transitions survive removal and surface as phantom objects—thin, wall-colored slabs that a downstream verifier confidently misreads (e.g., as wall-mounted keyboards or handrails, Sec. 5.3). We therefore filter any cluster that is (i) a thin vertical planar band whose (ii) upward neighborhood in the *pre-removal* cloud terminates in a dense horizontal layer—i.e., the band hangs from a real ceiling. The topmost-layer condition prevents false positives on window sills and wall remnants, which do not have a ceiling directly above. The filter reduced the showroom scene from 45 to 38 clusters and the robot hall from 88 to 81 while leaving every retained cluster byte-identical, so previously computed verification results remain valid. Horizontal and sloped ceiling patches are out of the filter’s scope and are discussed as a residual limitation in Sec. 7.

4.3 Explicit Modeling and Provisional Classification

Each cluster is summarized by an explicit model: axis extents and centroid, yaw from the principal component of the horizontal point distribution, dominant color, and point count. These attributes are deterministic functions of the cluster; the backbone’s determinism claim covers geometry and these explicit attributes only—the provisional label below is stochastic and excluded from it. A provisional semantic label is then attached by Uni3D open-vocabulary classification over a 30-class industrial vocabulary. The provisional label is explicitly *not trusted*: our own vocabulary ablation (Sec. 5.6) shows that narrowing the vocabulary to the deployment domain *reduces* accuracy (66.7% $\rightarrow$ 33.3%), indicating an embedding-level domain gap that cannot be fixed by class-list curation. The provisional label serves only as VLM context and as a fallback name; every label a robot may act on must pass the verification stage below.

4.4 Multi-View VLM Verification with Automatic Escalation

Verification renders each object from four azimuths (0° , 90° , 180° , 270°) and submits each view independently to a VLM (`claude-sonnet-4-6`) under a forced tool-use schema: the model must return a type, a confidence, and a stated reason, with the object’s measured dimensions and provisional label provided as context. Free-text answers are impossible by construction, and prompts, decoding parameters, and model version are recorded with every result.

Per-view votes are combined by confidence-weighted late fusion into a fused type and a *vote share*. Three outcomes follow. If the votes are unanimous, the object is **verified**; unanimity is empirically a strong signal (94.4–100% precision on the two scenes’ confirmed ground truth, Sec. 5.3), although Sec. 5.3 also documents its failure modes. If the vote splits, the object *escalates automatically*: eight additional images—four diagonal azimuths and four zoomed close-ups—are classified, and the fusion is recomputed over all twelve votes. If the post-escalation share reaches 0.6 the object becomes **verified_escalated**;

chair_040 — split base vote → automatic escalation → fused 'chair' (share 0.677, verified_escalated)

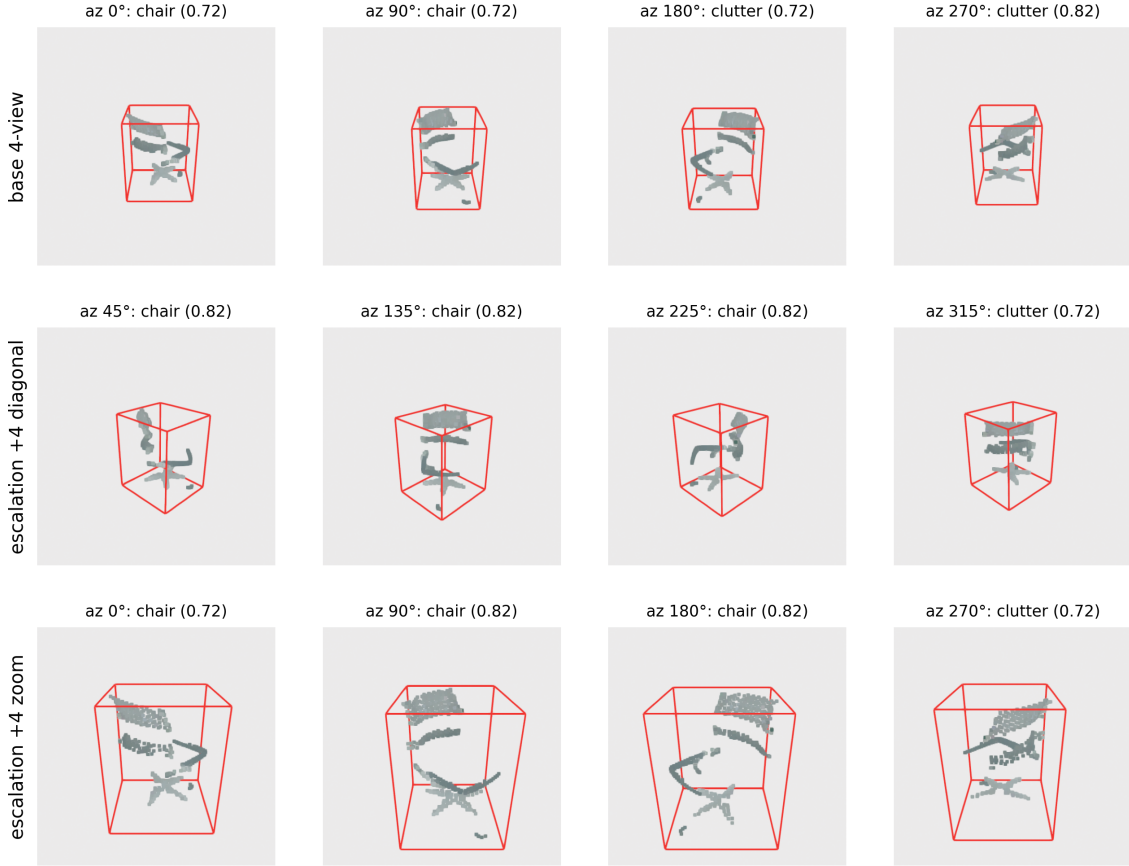


Figure 3: Multi-view verification and automatic escalation, with per-view votes, fused share, and the three terminal statuses.

otherwise it remains **unverified**. Crucially, unresolved objects are not hidden, discarded, or resolved by a human: they stay in the graph with their failed candidate label attached, and the query interface (Sec. 4.7) is responsible for making their status impossible to ignore. Implicit robot-facing attributes (movability, openability, key-object status) are inferred once at fusion time from the winning view set.

Escalation bounds cost: only split votes pay for the wider view set (trigger rates 51–58%, Sec. 5.3), and per-call cost is $\sim \$0.009$, so a scene verifies for a few dollars (Sec. 7).

4.5 Object Identity Matching

The update problem is an *identity* problem before it is a geometry problem: a robot that plans against node IDs needs yesterday’s chair to be today’s chair. Our policy is **mint-once**: a node ID is minted when an object first enters the graph—derived from a fingerprint of its birth-time anchor (map, position, extent)—and never changes afterwards. The fingerprint is a candidate-retrieval key, not the identity itself; after minting, matching alone carries identity forward.

Matching between a new scan’s records and existing nodes runs in two passes:

1. **Spatial pass.** A record matches a node if centroids agree within 0.5 m and each extent agrees within 30% (or 0.15 m for small objects). Matches update pose and extents in place (**unchanged** / **updated**).
2. **Move pass.** Remaining records match remaining nodes of the *same type* with consistent dimensions within a 6 m search radius, producing **moved** nodes with preserved IDs. Generic types (**clutter**, **unknown**) are excluded from this pass: a generic label carries too little identity for displacement claims, and our real two-epoch data showed that admitting them fabricates 2–5 m “moves” between unrelated piles (Sec. 5.5).

Records that match nothing are minted as new nodes (**inserted**); nodes that no record claims are marked **absent** but retained with their full history, so an object that returns later resumes its original identity. Sec. 5.5 evaluates matching accuracy, identity-switch rate, and false-new/false-absent rates, including the occlusion level at which matching breaks (40% crop).

4.6 Confidence-Gated Graph Upsert

Each scan epoch applies its matched record set to the knowledge graph as a node-level upsert—no rebuild. Every node carries its verification status, provenance (per-view votes, fused share, escalation flag, model version), a per-revision history of every change, and **firstSeen/lastSeen** timestamps. Edges (spatial relations—**isNextTo**, **isOn**, **isAboveOf**—recomputed from explicit models) are scoped to the map and revision, and organized by the place layer: **isNextTo** is kept only between objects of the same place (proximity across a place boundary is expressed at the place level as **isAdjacentTo**), vertical relations must pass a rotated-footprint overlap test (a separating-axis check that discards the long-range stacked relations oversized merged clusters otherwise produce), and each place designates a key object (the VLM’s implicit **isKeyObject** among verified members, falling back to the most confident specific-type verified member; objects refuted by owner ground-truth feedback are excluded from designation while remaining in the graph as annotated errors). On the T3 epoch this scoping removes 24 of 49 raw proximity edges—long cross-place spokes centered on the largest cluster—leaving 25 intra-place edges over 11 place adjacencies; Fig. 4 shows the resulting map. The result is exported both as a canonical JSON graph and as mapId-scoped Neo4j Cypher, so multiple sites and epochs coexist in one database. Because unchanged objects reproduce byte-identical clusters (Sec. 4.2), their verification results are carried over by content hash and a repeat scan costs *zero* VLM calls; across-epoch scans pay only for changed clusters (3.1% of full cost in our measurement, Sec. 5.5).

4.7 Query Interface with Structural Gating

The graph is queried through an LLM that receives a serialized view of the relevant nodes. The naive design—instructing the model not to assert unverified labels—fails in practice: in our benchmark the instructed model still asserted trap labels in 3 of 4 cases (Sec. 5.6). We therefore *gate by construction, not exhortation*: in the query-time view, every unverified node’s type is demoted to **unknown**, and the failed candidate label is moved to a separate **failedCandidateType** field whose semantics (“a classifier guess that did not survive verification”) are stated in the schema, not in behavioral instructions. Fig. 6 contrasts the two record states as the query interface sees them. With the gate in the data rather than the prompt, the model cannot assert an unverified label without visibly contradicting its input: the four unverified-label traps are blocked in the gating-style ablation (4/4, Sec. 5.6), while in the end-to-end benchmark one count-phrased trap still leaks through the quarantine field itself (3/4, Sec. 5.4). Traps whose label was wrongly *verified* lie beyond any status-based gate—a bound we quantify in Sec. 5.4.

4.8 Object-Grounded Place Layer

Robot tasking refers to places (“the docking area”) as often as to objects. Following the TOSM place schema of DK-SMF [5]—boundary polygon, containment, and purpose—we add a place layer grounded in the verified object layer. Place regions are obtained automatically from the robot-facing occupancy grid map (0.05m—generated from the scan here, and equally the exploration SLAM map on deployment), following DK-SMF’s recipe: the room interior delimited by the mapped outer boundary is partitioned by SLIC superpixel segmentation [16] with the number of superpixels as the governing parameter (7 requested here), fragments below an area threshold (1.5 m²) are discarded, and each region’s centroid is the moment-based center of its cells (cf. grid-map room segmentation [17, 18]). Each object’s **isInsideOf** resolves to the region containing its anchor cell. Each place is then summarized by its *ring code*: the sequence of member objects sorted by bearing around the place centroid—restricted to verified-tier objects, so the confidence gate propagates into place semantics—and a single LLM call names the place from its ring code. Because the segmentation depends only on the map, region identity is stable across scan epochs, and the same regions re-derive names that track the room’s actual use (the machine bay: exhibition-era **machine_access_area** → storage-era **clutter_zone** → test-era **machine_operator_station**; Sec. 5.5), demonstrating that place semantics stay current without manual re-annotation. Fig. 7 shows the resulting T3 place layer. The robot layer of TOSM is out of scope here.

place-scoped TOSM relation map (T3, rev 4): isInsideOf containment, intra-place object edges, place adjacency, and per-place key objects

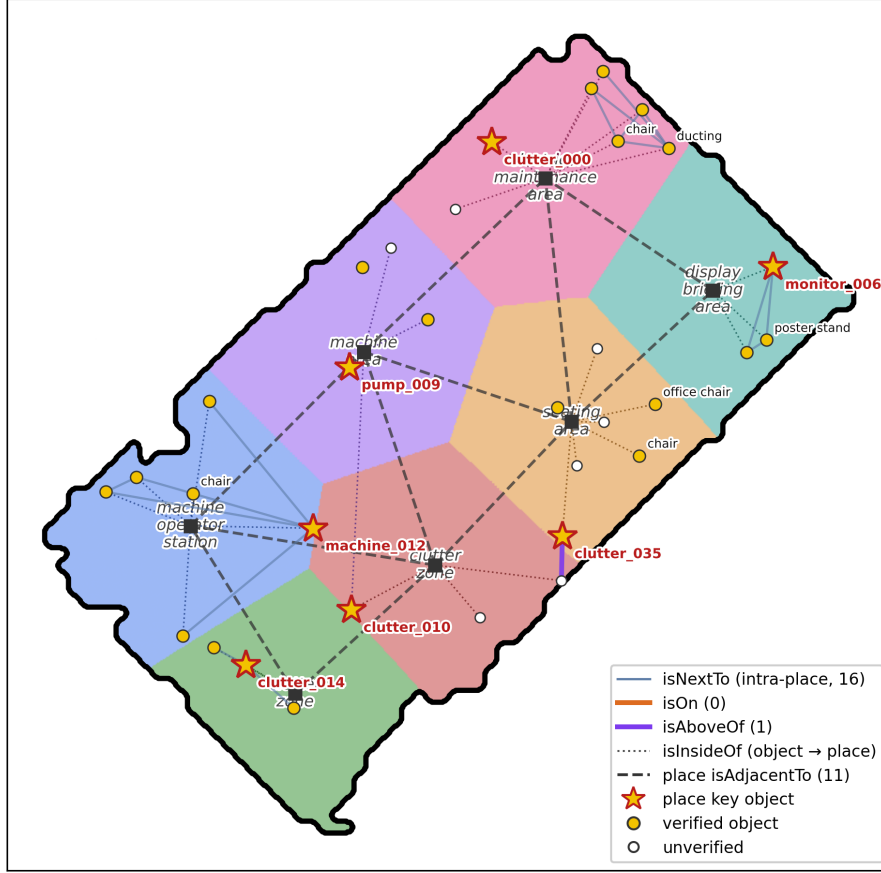


Figure 4: Place-scoped TOSM relation map (T3) over the place layer of Fig. 7 (pastel): dotted object→place isInsideOf containment, intra-place isNextTo edges, vertical isOn/isAboveOf relations, dashed place isAdjacentTo edges between region centroids, and per-place key objects (named stars), shown after the owner-feedback revision (Sec. 5.3): refuted phantoms (a verified keyboard, staircase, and book that do not exist in the room) have been removed from the graph, the poster stand carries its owner-corrected label, and the recovered TV is the key object of its region.

5 Experiments

5.1 Setup

Scenes and epochs. All data comes from BLK360 scans of two industrial indoor environments that are, in fact, one room observed across its working life—a property we exploit deliberately. The *showroom* epoch (June 2026) captures the room in an exhibition configuration, scanned manually with the door open; its semantic target is the wall-removed export (38 objects after remnant filtering). The *robot hall* epoch (July 2026) captures the same room reorganized as a robot test hall, scanned by autonomous visibility-driven exploration (81 objects, room-scoped to 55). A third epoch (late July 2026) adds a controlled-change re-scan at 99.6% coverage (Sec. 5.5). Because the two primary epochs differ in layout, furnishing, and scan modality, they function as independent scenes for per-epoch evaluation (Secs. 5.2–5.4) while enabling a real multi-epoch update study (Sec. 5.5). Scene statistics are given in Table 3.

Ground truth. Ground truth was established in two passes: a render-based audit of every cluster, followed by confirmation from the room’s owner. Labels are graded *confirmed* (owner-verified), *plausible*, or *unknown*; accuracy is reported against confirmed GT (with lenient variants including plausible where stated), unknown objects are excluded, and structure artifacts are tracked separately. Two showroom sets must be distinguished: Table 3 counts the 38-object reproducible det set, while the verification study (Sec. 5.3) uses the original 57-cluster decomposition of the same scan, of which 50 clusters are

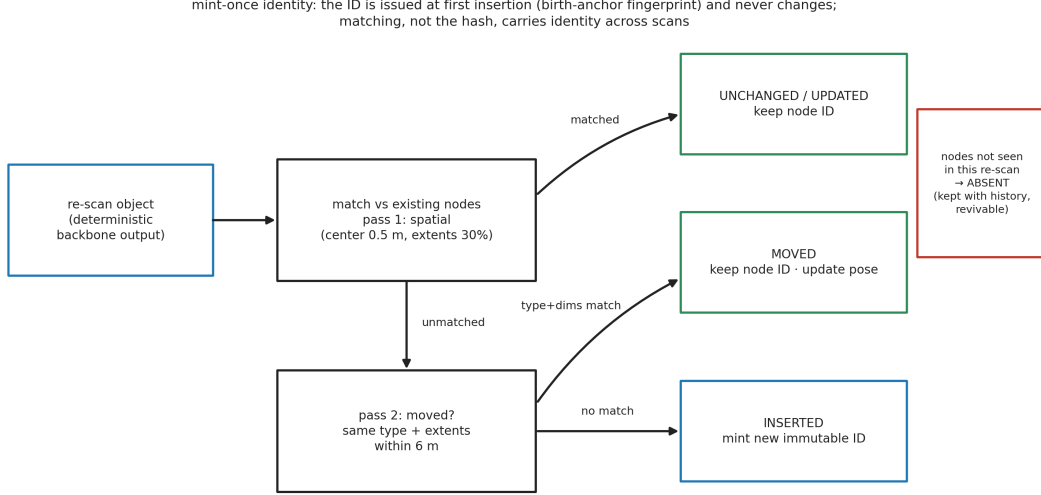


Figure 5: Identity matching and confidence-gated upsert: mint-once IDs, two-pass matching, and the five node outcomes across a re-scan.

(a) verified object record

Symbolic

type	control panel
name	control_panel_007
verified	true (VLM 4-view unanimous)
voteShare	1.0

Explicit

pose	x -2.24 y -3.54 z 1.00 m (map)
theta	3.03 rad
dims	1.87 x 0.43 x 2.83 m
color	brown
conf.	1.00

Implicit

keyObject	true
movable	false

(b) unverified record (gated)

Symbolic

type	unknown (gated; was 'keyboard')
name	keyboard_046
verified	false → demoted at query time
voteShare	0.281

Explicit

pose	x 8.59 y 4.22 z -0.58 m (map)
theta	1.67 rad
dims	0.58 x 0.06 x 0.51 m
color	brown
conf.	0.28

Implicit

keyObject	false
movable	true

Figure 6: TOSM semantic-object records (symbolic / explicit / implicit layers). (a) A verified record asserts its type. (b) An unverified record is structurally gated at query time: the type is demoted to **unknown** and the failed classifier guess is quarantined in **failedCandidateType**.

GT-scored after excluding 7 structure artifacts—yielding confirmed 36 / plausible 12 / unknown 2. For the room-scoped robot hall (55 objects), GT covers 51 clusters: confirmed 32 (lenient $n=42$) after owner corrections.

Metrics. Classification accuracy is top-1 against GT under a documented synonym-tolerant matching (e.g., *mobile manipulator* $\approx$ *robotic arm*). Verification quality is reported as per-tier precision (unanimous, escalated, unverified). For query evaluation, a *hallucination* is an unsupported atomic factual claim, and the rate is unsupported claims over all factual claims in the answer set. Update quality uses the metrics of Sec. 4.5: matching accuracy, identity-switch rate, false-new, false-absent, and unchanged stability.

Controls. All query conditions share the same LLM (claude-sonnet-4-6), system prompt, query set, decoding parameters, retry policy, and forced answer schema; conditions differ *only* in the knowledge view they receive. The LLM-only condition receives the same scene’s Stage-A object dump, so it is not information-starved—it is verification-starved.

object-grounded place layer (T3): SLIC superpixel regions over the nav-stack grid map;
LLM-derived names from verified-object ring codes

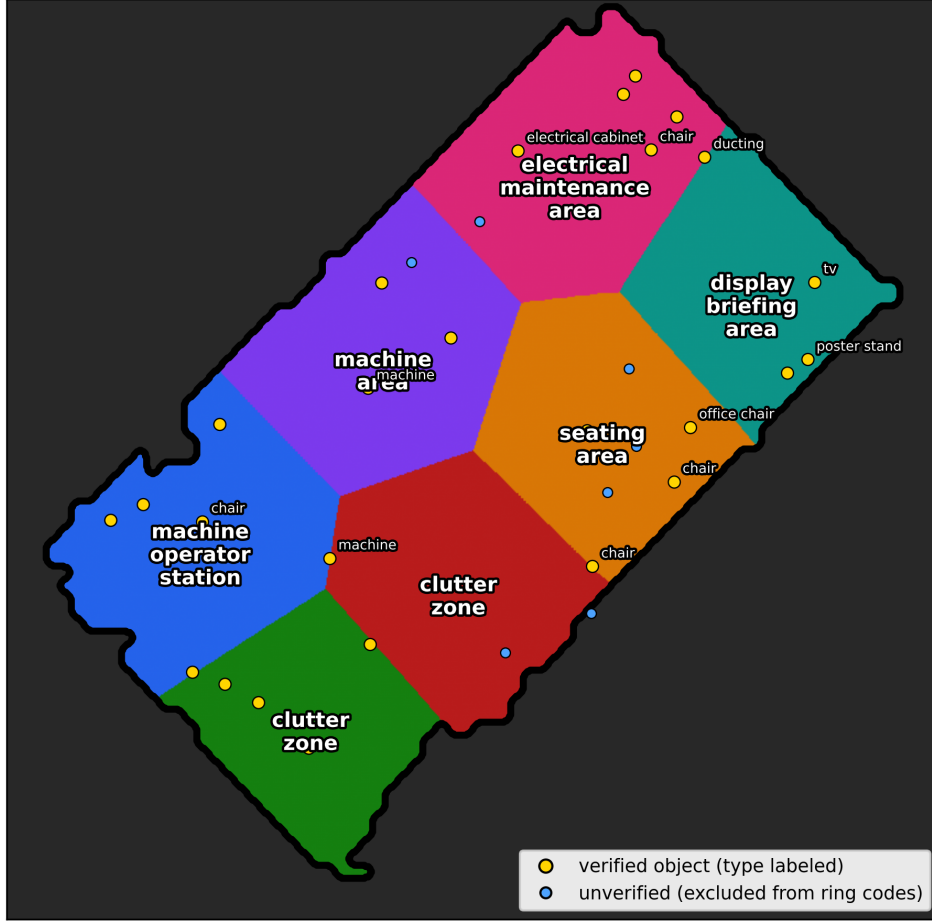


Figure 7: Object-grounded place layer at T3. SLIC superpixel regions [16] tile the room interior delimited by the nav-stack grid map’s outer boundary and carry LLM-derived names computed from verified-object ring codes; unverified objects (blue markers) do not contribute, so the confidence gate propagates into place semantics. Shown after the owner-feedback revision (Sec. 5.3): with the phantoms refuted, the region once named after a phantom staircase renames itself `display_briefing_area` around the recovered TV and poster stand.

5.2 Segmentation: Repeatability, Filtering, and Sensitivity

Repeatability. Running the full pipeline twice from the same E57—including ten-plane structure removal—reproduces every cluster byte-identically on both scenes (hash-verified). This is the property the incremental machinery builds on: identical clusters mean carryable verification results and attributable diffs.

Remnant filter. The structure-remnant filter of Sec. 4.2 removed all fourteen owner-confirmed ceiling-soffit phantoms (seven per scene: showroom 45→38, robot hall 88→81) while leaving every retained cluster unchanged. Before the filter, these phantoms did not merely inflate the object count: they produced confidently verified fictions (keyboard panels, handrails) that survived multi-view voting, since every view showed the same wall-colored slab. Detecting structure remnants *geometrically*, upstream of semantics, proved more reliable than asking the verifier to doubt itself (cf. the pose-context ablation, Sec. 5.7).

Parameter sensitivity. A 36-combination sweep (giant-split footprint $\times$ DBSCAN $\varepsilon \times$ minimum points) on both scenes shows a stable plateau at footprint 2.0–2.5 m: instance recall degrades at 3.0 m as neighbors merge (recall 0.929) and at 1.5 m as equipment shatters, while ε /min-point extremes collapse

Table 3: Scene statistics (det sets, after structure-remnant filtering). T1 semantic processing starts from a manually wall-removed export (4.9M points) of the 46.3M-point raw scan.

	Showroom (T1)	Robot hall (T2)	Re-scan (T3)
Scan mode	manual, 1 setup	exploration	exploration (SOTA), 6 setups
Raw points	46.3M	68.6M	70.2M
Clean cloud (post-removal)	—	267K	294K
Objects (filtered)	38	81	79
Room-scoped	38	55	41
Removed planes	10	10	10

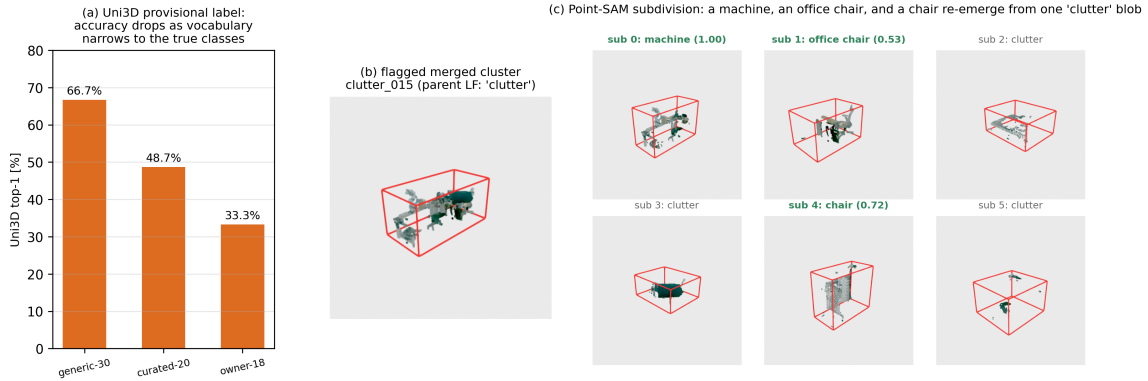


Figure 8: Learned components used as bounded tools. (a) The Uni3D provisional label degrades as the vocabulary narrows to the deployment’s true classes (Sec. 5.7)—why it is treated as untrusted context. (b,c) Conditional Point-SAM refinement of a flagged merged cluster: a machine, an office chair, and a chair re-emerge from a single `clutter` blob, each re-verified independently.

recall to 0.21. The operating point (2.5 m / 0.30 / 100) sits inside the plateau rather than on a peak. One caveat is circularity: the sweep’s reference instances are the audited operating-point decomposition, so the sweep measures stability around the operating point, not global optimality.

Conditional Point-SAM refinement. Under a pre-registered rule (adopt iff $\geq 50\%$ of merged-cluster failures separate correctly *and* label accuracy improves ≥ 5 points), Point-SAM separated 3/6 flagged merged clusters and sub-cluster labels reached 9/11 versus 5/11 for their parents (+36.4 points, Fig. 8), so refinement is adopted—but only for *flagged merged candidates*, since automatic candidate selection also produced 2/6 false positives that fragment single objects.

The T3 epoch supplied a stress case: at 99.6% coverage, inter-object gaps close and DBSCAN chains 120K points (41% of the clean cloud) into one 10.4×9.6 m mega-cluster that survives the giant-split. Refinement recovers little of the resulting diff inflation—Point-SAM ($K=7$ prompts) re-associates 4 of 43 absent nodes, tighter re-splitting ($\varepsilon=0.12$) 5 of 43—which is itself informative: $\sim 88\%$ of that epoch’s absences reflect genuine scene change, not segmentation artifacts (Sec. 5.5).

Backbone selection. The DBSCAN + Uni3D backbone was chosen against learned alternatives on a wall-removed industrial scene (122.7K points), summarized in Table 4. The metrics are deliberate proxies (no point-level ground truth exists for this scene): point coverage, the fraction of objects the open-vocabulary classifier cannot name (a boundary-quality proxy), and label diversity. ScanNet-pretrained SPFormer hallucinates furniture on industrial content (67 of 88 instances “chair”) at half the coverage; PTv3’s closed-set semantic head labels 46% of a scene *whose walls were already removed* as `wall` — a closed set cannot abstain; and PointCLIP V2 collapses to a single class. The geometric backbone wins not by being smarter but by refusing to guess: its failure mode is unlabeled coverage, which the verification and owner layers are built to absorb. Point-SAM enters only as the conditional refiner of Sec. 5.2.

Diffuseness-gated re-splitting. Two T3 clusters that survived every upstream filter are *diffuse webs*: 95K and 13K points spread over 80 and 12m^2 at a 2D occupancy fill below 0.3, versus 0.5–0.85 for every real object in the scene — a clean geometric signature, but one density clustering cannot act on,

Table 4: Segmentation×classification backbone comparison on a wall-removed industrial TLS scene (proxy metrics; no point-level GT).

Pipeline	Coverage	Unnameable	Classes	Dominant failure
DBSCAN + Uni3D	98.3%	23.3%	13	unlabeled coverage
SPFormer + Uni3D	51.1%	45.9%	9	furniture bias (chair 67/88)
PTv3 (semantic)	—	—	17/20 fire	46% “wall” after removal
DBSCAN + PointCLIP V2	98.3%	—	1–7	collapse (pipe 23/30)

Table 5: Verification accuracy against confirmed GT, and per-tier precision. Accuracy rows are over confirmed GT ($n=36/32$); trigger rates and tier counts are over all GT-scored clusters (50 showroom / 51 robot hall), so, e.g., 21 unanimous + 29 split = 50.

	Showroom ($n=36$)		Robot hall ($n=32$)	
	base-4	escalated	base-4	escalated
Accuracy	75.0%	86.1%	78.1%	84.4%
Recovered / regressed	5 / 1		3 / 0	
Escalation trigger	58% (29/50)		51% (26/51)	
Unanimous precision	100% (21/21)		94.4% (of 25)	
Escalated precision	100%		88.9%	
Unverified precision	50%		40%	

because the webs are connected at ~ 1.8 cm point spacing at any usable ε . A two-stage re-split resolves them by *cause* instead of size: points hugging the exploration-grid wall line (inside the *room polygon*, so wall-flush furniture survives the test that free-space masks would fail) are structure noise ($\sim 70\%$ of the 94K-point residual); out-of-room bleed-through accounts for another 7%; the remainder is re-clustered by occupancy-core erosion and, for compounds, split at the support plane detected as the z -histogram peak 0.5–0.95 m above floor. The owner’s ground truth for the largest compound — “a robot arm on a desk, part of a mobile robot, wall noise” — is recovered exactly: the arm (consensus-verified), the desk (slab plus under-desk storage), and the partially scanned mobile robot, plus three columns and a plant. Applied to a 6.2 m wall-flush cluster that verification had confidently labeled **electrical cabinet** (0.83), the same machinery — wall-band strip, support plane, then density-valley cuts along the wall in the panel band — yields *exactly* the owner’s inventory: ten monitors (five columns of stacked pairs), the desk beneath them, and, from the off-wall remainder, a wheelchair robot and a single table leg. The gate turns “too diffuse to be an object” from a discard decision into a decomposition trigger.

Structure-condition ensemble. Structure removal is itself a segmentation parameter: it amputates wall-flush objects along with the wall. Running the geometric pass under three removal conditions (none / ceiling only / full) and fusing the results recovered two multi-metre tables that exist in *no* single-condition segmentation of the production path — they appear only when walls are left in place — raising in-room recall by four real objects ($\sim 8\%$). The fusion also quantifies label fragility: only 10 of 37 re-found objects keep the same label across conditions, and cross-condition *voting* corrected nothing, because the verifier’s biases are correlated across conditions (both recovered tables were unanimously mis-verified as conveyor belts in every condition, while the room’s actual conveyor unit stayed **clutter** in every condition). The ensemble therefore ships as a recall tool plus a per-object **labelStability** attribute that prioritizes owner review — not as a label fuser. Figure 9 shows the wall-compound machinery end to end on the room’s monitor wall.

5.3 Verification: Fusion and Escalation

Escalation lifts confirmed-GT accuracy from 75.0% to 86.1% on the showroom and from 78.1% to 84.4% on the robot hall (lenient: 76.2% $\rightarrow$ 83.3%, $n=42$), recovering 5 and 3 split-vote objects while regressing 1 and 0 (Table 5; Fig. 11 shows the accuracy–cost trade-off at $\sim \$0.009$ /call; escalation adds 232 and 208 calls per scene). The tier structure justifies gating: unanimous and escalated tiers reach 88.9–100% precision, while the unverified residue is 40–50% — exactly the content the query interface must demote. A separate check evaluates the room-scoped T1 objects in the update-study frame (Sec. 5.5), where showroom GT is transferred to the re-decomposed clusters by bidirectional point overlap; on the resulting small confirmed

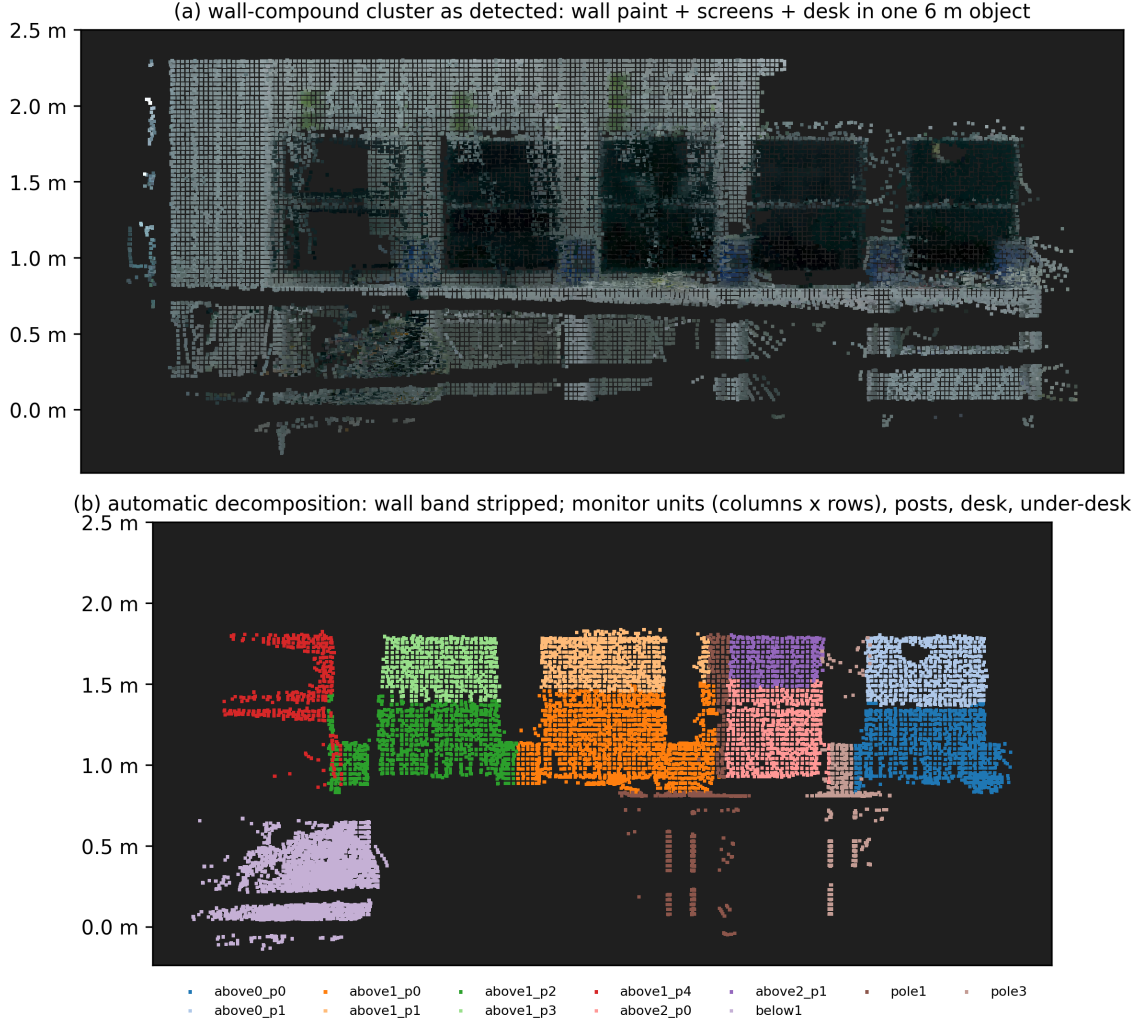


Figure 9: Automatic wall-compound decomposition (no owner input). (a) As detected: a single 6 m cluster mixing wall paint, ten dark monitor screens, support posts, and the desk — confidently verified as one “electrical cabinet” by the baseline pipeline. (b) After the wall-compound gate: the mapped wall band is stripped, the support plane separates the desk, u-density valleys cut the panel bank into columns, a conditional z-cut splits stacked rows, and a square-up pass re-assigns boundary points — yielding nine of the ten owner-confirmed monitor units plus posts, desk, and under-desk storage.

set ($n=8$), verified-tier precision is 100% with 7/7 unanimity.

Verification yield tracks scan quality: the manual scan leaves 47% of objects unverified versus 24% for exploration scans, and re-processing the T3 scan from a single setup (1/6 coverage) drops the verified tier from 75.6% to 56.6% on identical geometry. Coverage buys verifiability.

Error taxonomy. Four recurring error classes emerged (Fig. 12): (a) *structure-remnant phantoms*—soffit bands verified as keyboards; eliminated upstream by the remnant filter; (b) *unanimous-but-wrong*—a hanging chain unanimously verified as a door handle; an escalated keyboard the owner refuted; partition fragments as chairs; at T3, a residual 7.4 m wall segment that survived the remnant filter was escalation-verified as a staircase—owner-refuted (“just the wall”)—and its phantom label then propagated into a place name (`staircase_landing`, Sec. 4.8), showing that place naming is verifier-bounded too and that wall-scale remnants can slip past a filter tuned for ceiling-soffit bands (cf. Sec. 7). The same propagation reaches the relation layer: a keyboard verified at full confidence in a room that contains no keyboard (owner-refuted) had been auto-designated its place’s key object before the refutation demoted it—key-object designation therefore honors owner feedback (Sec. 4.6). A third refutation is the mirror image: the room’s large wall-mounted TV was geometrically detected (`monitor_006`, 2.5×1.8 m) yet left **unverified** at 0.26 confidence, while a small co-located cluster was escalation-verified as a fire extinguisher (0.72) the

owner refuted—the salient real object failed the gate a phantom passed. Unanimity is strong evidence, not proof, and the gate inherits the verifier’s blind spots; (c) *clutter absorption*—a disassembled robot, a poster stand, and a low-reflectance floor fan (674 points) each unanimously verified as **clutter**: geometrically detected, semantically flattened; (d) *render ambiguity* → **unverified**, the intended failure mode. Classes (b) and (c) motivate reporting verification as *tiered evidence* rather than truth, and drive the structural gating design.

Owner ground-truth audit (T3). The room occupant exhaustively labeled all 41 T3 clusters, giving a second, fully independent ground truth beyond the showroom scoring sets. Of the 31 verified-tier clusters, 6 were correct, 13 were real objects flattened to **clutter**, 6 were structure-noise phantoms, 2 carried a wrong specific label, and 3 were multi-object merges (one unanswered). The absorption cause is now identifiable: the room is a robotics testbed (mobile robots, manipulators, charging stations, a disinfection robot), and none of these classes exist in the classifier vocabulary. The unverified tier behaved as designed in five structure/fixture cases but also withheld three real objects, including the wall TV of Sec. 5.3. Three audit-driven mitigations follow. (i) *Hybrid structure filter*: registering every cluster against the exploration-scan occupancy grid — an independent, drift-free wall prior — and combining a wall-line hug test with local (per-metre-segment) sheet thickness, a full-height band rule ($z\text{-span} > 2.5\text{ m}$), a ceiling band, and a room-polygon outside test removes structure noise at 12/12 recall with 3 false positives (thin wall-flush real objects: a cabinet leg, a poster stand). Crucially, the wall-flush monitor bank survives: its wall-hug fraction is 99% (the grid map draws it as wall), but its 1.8 m height and per-segment depth separate it from the 3.2 m floor-to-ceiling wall band that plane removal missed. All seven verified structure phantoms — including the staircase band and a fluorescent fixture verified as a keyboard at 3.3 m — are eliminated upstream. (ii) *Recovery re-verification*: re-asking the verifier with **clutter** treated as *unidentified* plus a candidate shortlist recovered 4 specific types at 4/5 precision with 12 honest keeps; injecting site vocabulary (robot classes) as candidates doubled recovery (9 correct) but halved precision (8 wrong) through candidate anchoring — the TV became a “conveyor belt” when the shortlist ranked it first. (iii) *Agreement gating*: adopting a recovered label only when both passes agree, or when a pass agrees with the cluster’s own base label, yields five adoptions at 5/5 precision (two chairs, an office chair, a machine, and the TV) and resurrects no owner-refuted phantom. Domain knowledge helps recall; only agreement makes it safe. Fig. 10 shows the filter’s effect on the T3 epoch. Regression on the other epochs (no ground truth, manual inspection) flags 8/50 T1 and 18/55 T2 clusters — dominated by the documented soffit phantoms (keyboards, chairs at 2.4 m, a stair at 3.4 m) and a cluster the classifier itself had named **wall** — while every door, chair, machine, elevator, and wall-mounted control panel survives; the one systematic casualty is a closed-door fragment lying in the wall plane, consistent with the door-detection asymmetry of Sec. 7. Finally, refutations, corrections, and adopted labels are applied as a fourth KG revision through the same upsert history mechanism as a rescan: eight phantoms leave the graph, the poster stand is corrected in place, five recovered types land, the TV becomes its region’s key object, and the relation layer sheds nine noise-anchored edges (25 to 16 intra-place). Re-deriving ring codes and names from the revised graph heals the place layer end-to-end: the phantom-derived **staircase_landing** becomes **display_briefing_area** (TV + poster stand), the keyboard-derived **workstation_area** becomes **machine_area**, and the seating region’s recovered chairs turn **cluttered_seating_area** into **seating_area** — owner feedback propagates through all three layers of the model: object, relation, and place.

Owner feedback at scale: bidirectional, corrective, and time-variant. Extending the audit loop through nine further KG revisions shows owner feedback operating in all four directions the model supports. *Restoration*: an object the owner had refuted as absent was later declared present again (the room changed between audits); a color-seeded carve at the owner-given corner recovered a 319-point red cylinder that two-pass verification typed **fire extinguisher** (0.72) — the original failure was segmentation (never proposed), not classification, and owner ground truth is itself *time-variant*, which is why feedback is recorded as highest-trust evidence with history rather than as immutable truth. *Refutation*: a raw-condition ensemble find was refuted as wall paint — the patches surround the room’s wall-mounted TV, whose glossy black screen returns almost no lidar, so the “object” was the halo around a hole. *Label correction*: both ensemble-recovered tables (unanimous “conveyor belt”) and a poster stand (tie-break “fire extinguisher”, 0.72) were corrected in place, each with the VLM’s verdict preserved in provenance — unanimity across passes *and* across structure conditions still inherits one model’s bias. *Attribute correction*: wall-remnant clutter was marked immovable in the implicit layer. Adjudication itself gained three inexpensive refinements along the way: canonical synonym matching (“robot_arm” vs

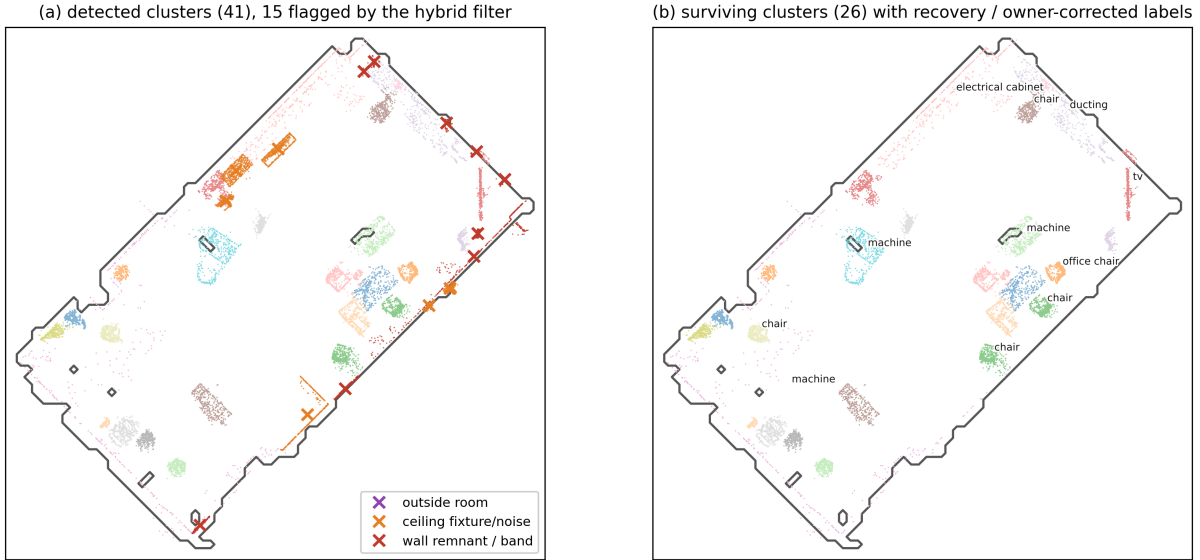


Figure 10: Hybrid structure-noise filter on the T3 epoch. (a) All 41 detected clusters; the 15 flagged ones are colored by rule — wall remnant/band (grid-map hug $\times$ thin vertical sheet, or long full-height band), ceiling fixture/noise, outside the room boundary. (b) The 26 survivors with recovery / owner-corrected labels. The wall-flush monitor bank (labeled `tv`) and wall-mounted equipment survive; residual merge labels (`electrical cabinet`, `ducting`) are retained pipeline output, not corrections.

Table 6: Query benchmark, four conditions (accuracy / hallucination rate / traps blocked).

	LLM-only	Ungated	Verified-only	Gated
Showroom	52.1% / 20.8% / 2/4	81.2% / 18.8% / 0/4	58.3% / 16.7% / 4/4	83.3% / 16.7% / 3/4
Robot hall	50.0% / 37.5% / 4/4	56.2% / 43.8% / 0/4	46.9% / 40.6% / 2/4	59.4% / 40.6% / 2/4

“robot manipulator” is agreement, not disagreement), a focused tie-break pass that re-asks with exactly the two disputed labels (resolving desk vs machine at 0.62), and a coarse three-level height attribute (low/mid/high relative to the 2D-lidar scan plane and ceiling) injected as a hard physical constraint — the targeted successor to the pose-context defense of Sec. 5.7, and now a queryable implicit attribute of every node.

5.4 Knowledge-Grounded Query Evaluation

We evaluate four conditions over identically constructed per-scene query sets—48 items on the showroom, 32 on the robot hall (symbolic, explicit, scene-level, and trap questions): (1) *LLM-only* (Stage-A dump, no verification), (2) *ungated KG*, (3) *verified-only KG* (unverified nodes removed), and (4) *gated KG* (unverified retained but structurally demoted, Sec. 4.7). Each scene carries four trap questions about objects that do not exist as labeled. On the showroom all four traps target *unverified-label* phantoms (keyboard, ladder); on the robot hall two target an unverified-label phantom (refrigerator) and two an *escalated-verified* phantom (keyboard)—the latter probing the gate’s verifier-quality bound. A correct system must decline to assert the trap labels. Note that LLM-only can “block” traps trivially: with no map it tends to deny that specific objects exist at all (4/4 on the robot hall, 2/4 on the showroom), so trap blocking is diagnostic only for the grounded conditions.

On the showroom (Table 6), grounding alone (ungated) lifts accuracy from 52.1% to 81.2% but blocks no traps; removing unverified content (verified-only) blocks all four traps at a steep recall price (12 abstentions, 58.3%); the gated condition is the balance point—83.3% with 3/4 traps blocked. The two leaks are instructive and mechanistically distinct. On the showroom, one *count*-phrased trap slipped through the quarantine itself: the model tallied a demoted node by reading its `failedCandidateType` field, even though it correctly denied the same object in yes/no phrasing. On the robot hall, the keyboard

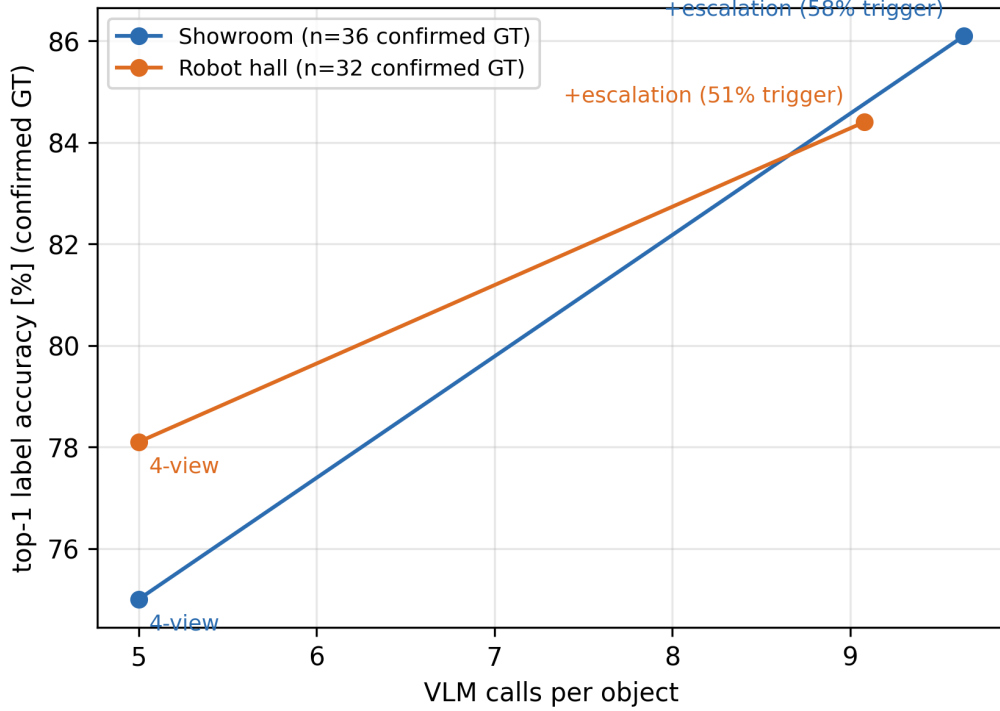


Figure 11: Escalation accuracy–cost curves for both scenes.

phantom is *escalated-verified*, so it passes any status-based gate—gating is bounded by verifier quality; we report both as properties, not anomalies. The robot hall reproduces the ordering at lower absolute levels because its end-to-end label errors (banner→refrigerator, robot→clutter) propagate into every grounded condition—an honest end-to-end number rather than a curated one.

5.5 Incremental Update Evaluation

Synthetic scenarios. Controlled edits of the room-scoped robot hall isolate matcher behavior (Fig. 14): (A) *unchanged repeat*: 55/55 nodes carried, zero VLM calls, zero identity switches; (B) *move*: 1/1 moved with ID preserved; (C) *remove+insert*: exact absent/insert classification; (D) *occlusion*: at 20% crop both affected objects still match (as updated, labels kept), at 40% matching fails into 2 false-absents + 2 false-news—the matcher’s quantified breaking point, and the motivation for overlap-based matching features in future work. On the combined-edit re-scan (move, insertion, and removal applied together), selective re-verification re-ran Stage B on only the two changed clusters: 26 of 822 VLM calls, \$0.23 versus \$7.37 for a full re-verification—3.1% of rebuild cost, and 0% for scenario A.

Real three-epoch study. The three epochs of Table 3 form a real update sequence: T1 (manual, June) → T2 (exploration, early July) → T3 (99.6%-coverage exploration with a controlled-change protocol, late July). Cross-epoch registration is fully automatic (FPFH+RANSAC then ICP; 5.1 cm RMSE for T1→T2, 4.2 cm for T3→T2). The T1→T2 diff records the room’s exhibition-to-test-hall reorganization (updated 4 / moved 3 / inserted 48 / absent 31; among the moves, a fire extinguisher displaced 3.0 m). This epoch also motivated the generic-type exclusion in the move pass: without it, clutter piles matched across 2–5 m as fictitious “moves” (9 candidates, 3 credible). The T2→T3 diff (matched 9 / moved 1 / inserted 31 / absent 45, Fig. 15) sits on top of measured natural drift: only 65% of in-room object geometry is mutually supported at 10 cm between the two scans two weeks apart—roughly a third of the room changed before any protocol was applied.

Against the controlled-change protocol, the system detected **2/2 physical changes at geometry level**: the moved chair is the epoch’s *only moved* node (0.64 m, ID preserved—a node matched continuously since T1), and the introduced floor fan was correctly **inserted** but typed **clutter** (error class (c) above), giving 1/2 at type level. Identity persistence is demonstrated across all three epochs: the fire extinguisher, pump, industrial robot, and several chairs retain their minted IDs through both re-

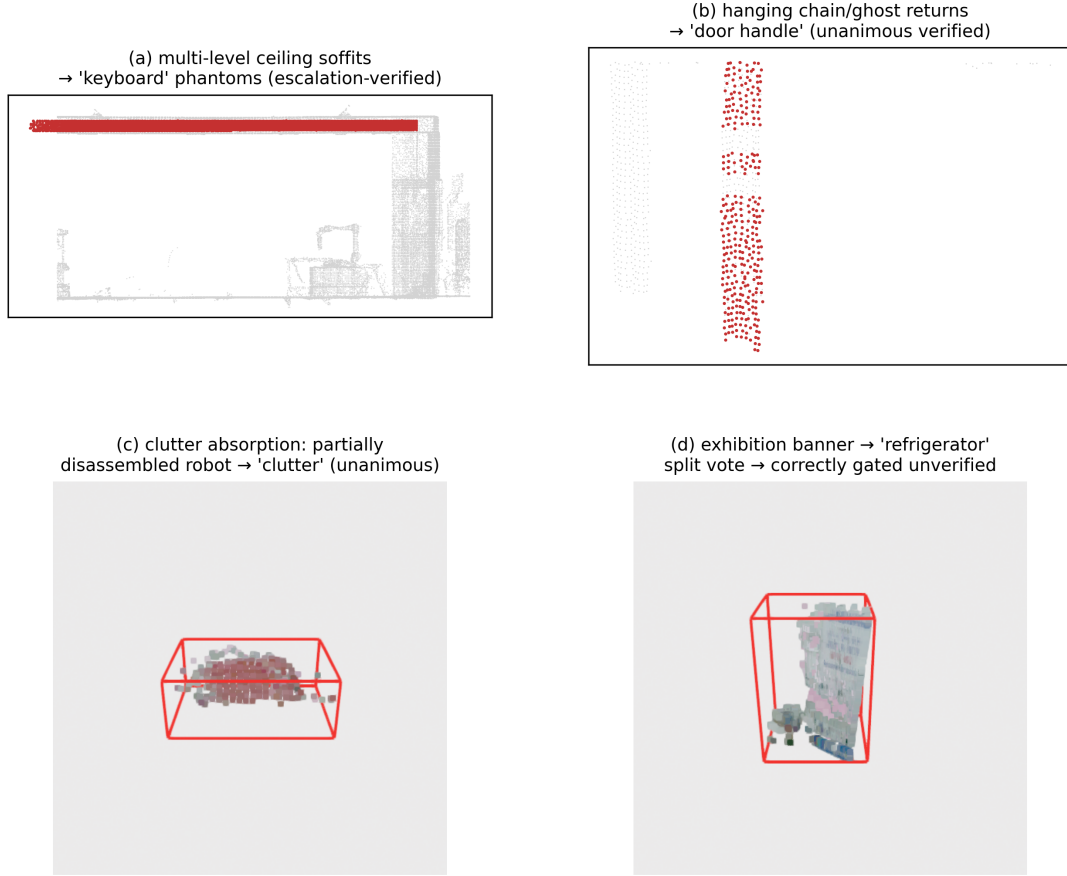


Figure 12: Verification error taxonomy: remnant phantom, unanimous-but-wrong, clutter absorption, render ambiguity.

organizations. The live graph has since accumulated thirteen revisions through the same mechanism — rescans, owner refutations/restorations/corrections, re-split decompositions, and finally a stale-node sweep in which eighteen earlier-epoch detections with no support in the current scan were marked absent; every transition, including the mistakes it corrects, is reconstructible from node histories. The place layer re-derives per-epoch names on the same SLIC regions (Sec. 4.8): exhibition-era `machine_access_area` and `operator_control_station` become clutter-dominated at T2 and `machine_operator_station/electrical_services_zone` at T3, at $\sim \$0.03$ per epoch.

5.6 End-to-End Re-validation against Accumulated Owner Ground Truth

After the corrective machinery of Secs. 5.2–5.5 stabilized, we re-ran the entire pipeline from the raw scan with no owner input and scored it against the live knowledge graph — by then a 46-object inventory embodying every owner restoration, refutation, and correction. Table 7 reports cumulative geometric recall as each automated layer is enabled. The verified path alone recovers 28%; diffuseness-gated re-splitting nearly doubles it; the wall-compound gate (which self-triggers on the monitor wall and decomposes it into nine of the ten owner-confirmed units) and the structure-condition ensemble bring automated recall to 85%. Type accuracy saturates far lower ($\sim 31\%$ of recalled objects) — the label bottleneck is the correlated verifier bias of Sec. 5.2, and closing it is precisely the owner layer’s role. For calibration within the lineage’s own reporting style, DK-SMF reports 78–81% detection accuracy against predefined ground-truth objects in its RGB-D environments [5]; our 85% is measured against an owner-audited inventory that includes partially scanned and wall-flush objects, though the sensing modality and ground-truth protocols differ. The re-run also reproduced the deterministic backbone’s stability end to end: the same diffuse webs, the same compound decompositions, and a first-pass verification of the fire extinguisher that had originally required owner restoration.

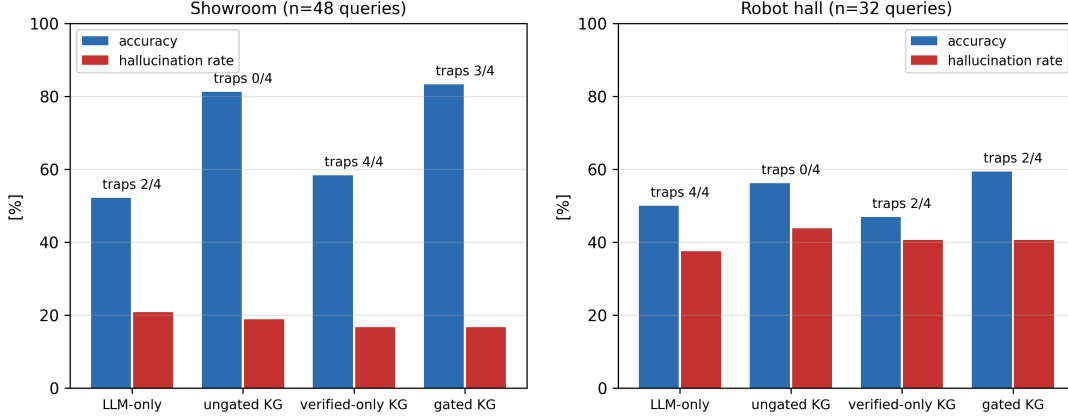


Figure 13: Four-condition query benchmark with trap-question panel.

Table 7: Cumulative automated recall of the full re-run against the owner-corrected 46-object graph (no owner input in the run itself).

Automated layers (cumulative)	GT recall	In-room precision
Verified path (Sec. 4.2–4.4)	13/46 (28%)	37%
+ diffuseness re-split	26/46 (57%)	53%
+ wall-compound gate	36/46 (78%)	60%
+ structure-condition ensemble	37/46 (80%)	49%
+ panel square-up	39/46 (85%)	51%

5.7 Ablation and Sensitivity

Escalation on/off is the base-4 column of Table 5: +11.1/+6.3 points. *Gating style*: instructing the model to respect verification status blocks 1/4 traps; structural demotion blocks 4/4 under identical prompts in this isolated comparison (in the separate end-to-end run of Table 6, one count-phrased trap leaked, 3/4)—still the single strongest argument that reliability must live in the data contract, not in instructions. *Pose context*: injecting mounting-height context as a targeted defense removed all three phantom ceiling keyboards in its pilot, but applying it to every object *reduced* escalated accuracy (86.1%→72.2%) and escalated precision (100%→72.7%)—a desk-height prior that helps ceilings hurts everything else, so it ships as a targeted defense only. *Uni3D vocabulary*: generic-30 66.7% > curated-20 48.7% > owner-18 33.3% (run-to-run variance ~5 points), confirming the embedding-level domain gap that motivates verification. *Verifier model* (Table 8): running the identical Stage-B protocol across five verifiers (two vendors, three tiers, two generations; reasoning disabled for parity) makes escalation an economic instrument — call volume tracks verifier decisiveness monotonically (247 to 383 calls as unanimous verifications fall from 26 to 9). Three findings. First, within a generation the smaller tier splits more votes and resolves fewer: its indecision inflates call volume by 14% (383 vs. 335 calls), clawing back part of its nominal 3× per-token price advantage — and what survives is paid for in the lowest type accuracy and the largest unverified residue (17 of 35 objects). Second, all three current-generation frontier models tie on type accuracy against the owner inventory (7/13), but differ two-fold in the calls they need to get there — the same-generation mid-tier (Sonnet 5) is the most decisive (26 unanimous, 6 escalations), and the premium tier’s extra deliberation (18 escalations) buys no additional accuracy here. Third, the ranking is generation-dominated, not vendor-dominated: both 2026-generation mid-tiers beat both older Claude tiers, and cross-vendor differences within the generation are smaller than the tier effects. Escalation volume is thus a label-free, online proxy for verifier decisiveness on the deployment’s own data — and a purchasing guide: pick the cheapest verifier whose escalation rate has saturated. The guide is robust where prices are not: a vendor repricing days before our runs moved one verifier’s end-to-end cost five-fold (Table 8, note 2), while the decisiveness ordering it would be applied to is measured on the deployment’s own data.

Segmentation parameters: see Sec. 5.2.

Controlled re-scan scenarios, room-scoped robot hall (55 objects; colored points = segmented clusters, thin boxes = explicit models)

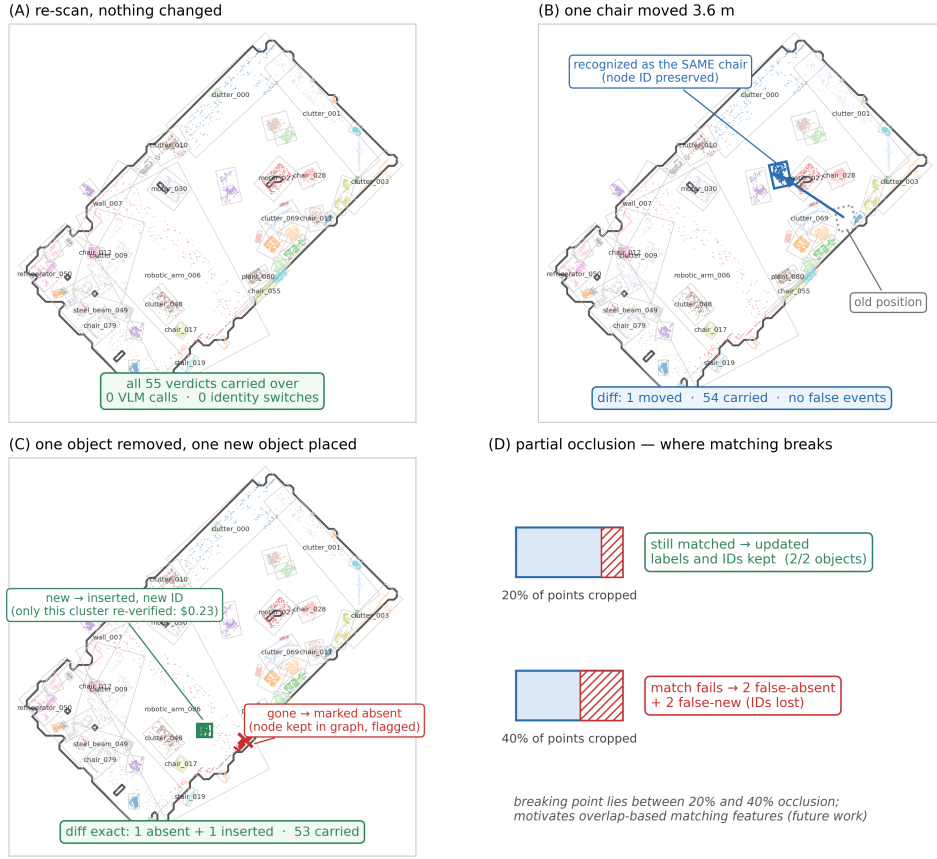


Figure 14: Controlled re-scan scenarios A–D (room-scoped robot hall, 55 objects), top-down view. Colored points are the segmented clusters, thin rotated rectangles their explicit-model footprints with node names, and the dark contour the test-room boundary; each edited object is annotated with the matcher’s decision. (A) A repeat scan with no changes carries all 55 verdicts at zero VLM cost. (B) A 3.6 m chair move is classified **moved** with its node ID preserved. (C) A removed object is marked **absent** (node kept and flagged) and a newly placed object is **inserted** under a new ID; only the insertion is re-verified (\$0.23). (D) Partial occlusion: both cropped objects still match at 20%, but at 40% matching fails into two false-absent plus two false-new pairs—the matcher’s quantified breaking point.

6 Application Demonstrations

This section demonstrates the framework in use. The semantic mission experiment of Sec. 6.1 is quantitative; the remaining demonstrations illustrate integration surfaces.

6.1 Semantic Mission Execution in Simulation

We close the loop from scan-derived knowledge to autonomous behavior. The TOSM robot layer is instantiated from the platform’s URDF and navigation configuration (footprint, drive type, sensor mounts, velocity limits) and ingested alongside objects and places, completing the three-element TOSM model. A *semantic mediator* resolves natural mission commands — `goto_place`, `goto_object`, `patrol`, `return_home` — against the knowledge graph: places resolve to free-space points at region centroids, objects to approach poses computed from the target’s footprint and the robot’s radius, with the verification gate deciding eligibility. A hybrid deliberative/reactive behavior tree (Fig. 16) executes the resolved goals through Nav2, with a reactive layer that preempts the mission on low battery or an interrupt and returns the robot to its home pose. The management console (Sec. 5.4) issues commands and edits the graph live; the mediator hot-reloads owner edits, so refuting a phantom in the UI changes robot behavior on the next command without a restart.

Experiments run in a Gazebo world generated from the same exploration occupancy grid that anchors

Table 8: Verifier-model comparison: identical Stage-B late fusion + escalation on the same 35 in-room clusters, scored against the owner-corrected graph.

Verifier	Verified	Esc.-resolved	Unverified	Type-correct ¹	Calls	Cost ²
Claude Sonnet 5	26	3	6	7/13	247	\$2.49
GPT-5.6 (luna)	21	7	7	7/13	287	\$0.11
Claude Opus 5	17	8	10	7/13	319	\$5.28
Claude Sonnet 4.6	15	6	14	6/13	335	\$3.00
Claude Haiku 4.5	9	9	17	5/13	383	\$1.13

¹ canonical-synonym type match on the clusters that geometrically match a ground-truth object. ² measured tokens (in/out: 0.63M/40K, 0.42M/19K, 0.79M/53K, 0.74M/52K, 0.84M/59K, row order) at vendor list rates of August 2026 (\$/Mtok in/out: 3/15, 0.20/1.20, 5/25, 3/15, 1/5). The GPT-5.6 row reflects the 2026-07-30 price reduction; at its launch rate (1/6) the same run costs \$0.54. Sonnet 5’s introductory rate (2/10) gives \$1.66.

Table 9: Phantom-goal mission waste in simulation (AMMR proxy). The same three commands are resolved against the knowledge graph before and after structure filtering + owner feedback (both with the verification gate active). Wasted path includes the return leg.

KG state	Phantom target (truth)	Outcome	Wasted path	Wasted time
pre-feedback	keyboard (fluorescent lamp)	driven	1.7 m	16.8 s
pre-feedback	staircase (bare wall)	false arrival	0.6 m	5.7 s
pre-feedback	book (wall pillar)	unreachable, aborted	0 m	22.2 s
filtered + owner feedback	all three	refused	0 m	~1 s

the place layer, so world, navigation map, and knowledge graph share one frame. The simulated vehicle is a digital proxy of the target platform itself: the AMMR’s footprint (1.25×0.8 m), lidar mount, and velocity limits from the robot layer are instantiated in the world, and the mediator derives approach poses and clearance checks from the same layer (circumscribed radius 0.74 m) — the knowledge that *describes* the robot is the knowledge that *drives* it. Over three repetitions of a three-mission battery (approach the TV, go to the seating area, return home), all nine missions succeeded (mean leg paths 4.6/5.4/1.8 m in 16.5/24.9/15.0 s). The phantom experiment (Table 9) sends the robot to three owner-refuted phantoms under two knowledge states. Against the pre-feedback graph, all three phantoms carry *verified* labels, so the confidence gate passes them — and the waste takes three distinct forms: a pointless drive to a fluorescent lamp; a *false arrival* at a 7.4 m “staircase” whose approach ring encloses the robot’s own position, so the mission reports success at a structure that does not exist; and 22 s of futile replanning toward a wall-pillar “book” whose approach pose the platform’s footprint cannot reach at all. Against the graph after structure filtering and owner feedback, the same commands are refused in about one second with zero motion. Verification gating alone cannot stop verified phantoms; it is the combination of gating, structure-noise removal, and owner feedback that makes the knowledge safe to act on. Fig. 17 shows a gated mission trajectory and an ungated phantom drive.

6.2 Knowledge-Grounded Dialogue

The gated query interface supports task-language interaction in which confidence is part of the answer, not metadata. Asked “is there a keyboard in the room?”, the system answers from the gated view: a verified keyboard is asserted with its location; an unverified one yields “an object at (x,y) was proposed as a keyboard by the classifier but did not pass verification”—an answer a robot can act on conservatively (route a person, lower approach speed) instead of a confident guess.

6.3 Simulation Twin Injection

Because every node carries an explicit model in the scan frame, the graph injects directly into a physics simulation. We reuse the digital-twin construction of our prior work [19] and add the semantic layer: each verified object spawns a labeled proxy (extent box or segmented cloud) in NVIDIA Isaac Sim, with verification status, provenance, height level, and label stability exposed in the scene inspector. Unverified objects spawn as neutral gray placeholders—the gate is visible in the twin. The knowledge graph is the *single* semantic store: the management console and the mission mediator read the same file live, and the Isaac stage is re-derived from it (in the shared map frame) whenever the graph changes, so an

owner edit propagates to the robot’s next goal, the console, and the twin from one write. The discipline is self-auditing — deriving the stage from the graph immediately *visualized* eighteen stale earlier-epoch nodes as orphan boxes that per-view export paths had silently hidden, prompting the cleanup revision of Sec. 5.5.

6.4 Manipulation Readiness

The implicit layer (movability, openability, key-object status) plus verified pose and extents provide the minimum contract a manipulation planner needs for coarse feasibility screening. We state this as an interface property; manipulation experiments are out of scope.

7 Discussion

Cost and latency. VLM verification is the only paid stage. Escalation is bounded by its trigger rate (51–58% of objects), and per-call cost is $\sim \$0.009$; full verification averages $\sim \$0.09$ per object ($\$3.63$ for a 38-cluster run and $\$7.37$ for an 86-cluster run, all calls included). The incremental design changes the cost model qualitatively: a repeat scan costs $\$0$ (hash carry-over) and a changed scene costs in proportion to its change (3.1% of rebuild in our measurement). Verification is offline; no VLM call sits on a robot’s control path.

Residual unverified content. On exploration-quality scans, 24% of objects remain unverified (47% on the manual scan; 43% on the 1/6-coverage ablation)—scan quality is the dominant driver of verification yield. The framework’s position is to expose this residue, not shrink it cosmetically: unverified nodes stay queryable with demoted types, and the gate is only as good as the verifier (Sec. 5.3’s unanimous-but-wrong cases; the escalated-verified keyboard trap that survives gating on the robot hall). Verifier-bounded gating is a property users must design around, and we report it as such.

Small low-reflectance objects. The clutter-absorption error class—disassembled robot, poster stand, and the controlled-change floor circulator all verified as `clutter`—shows a systematic blind spot: geometrically detected, semantically flattened. Change detection still works (the circulator was correctly inserted), but type-level recall for small dark appliances needs either close-range imagery or targeted re-rendering. The T3 owner audit (Sec. 5.3) adds a vocabulary dimension: when the deployment site contains object families absent from the classifier vocabulary (robots, in our testbed), absorption becomes the dominant error, and naively injecting site vocabulary as candidates trades precision for recall through candidate anchoring; agreement gating restores precision at reduced recall.

Structure remnants. The soffit filter removes vertical transition bands, but horizontal and sloped ceiling patches and content seen through openings survive structure removal (six residual cases in the robot hall). These are bounded, enumerable artifacts. The T3 audit shows the practical fix: a hybrid filter that registers clusters against the exploration-scan occupancy grid recovers all remaining wall, ceiling, and outside-boundary remnants (12/12) while keeping wall-flush furniture, at the cost of thin wall-flush objects (a poster stand) — geometry alone cannot separate a 0.1 m sheet standing at the wall from the wall itself. Structure removal has a second, opposite failure mode that the ensemble of Sec. 5.2 exposes: it *amputates* wall-flush objects (two multi-metre tables, a wall-mounted monitor bank) along with the wall. Both failure directions argue for treating removal as a varied condition to detect across, not a fixed preprocessing truth.

Correlated verifier bias. Cross-condition voting corrected no labels: when the verifier mislabels, it tends to mislabel the same way regardless of how the input was segmented (the recovered tables were “conveyor belts” under every structure condition; the actual conveyor was `clutter` under every one). Ensembling over *inputs* cannot decorrelate a bias that lives in the *model*; what the ensemble does yield is a calibrated instability signal (`labelStability`) that tells the owner where to look. Together with the time-variant ground truth of Sec. 5.3, this shapes our position: the owner is not a fallback annotator but the highest-trust evidence source in a permanently open loop, and the model’s job is to make each owner minute maximally productive — surfacing the unstable, preserving provenance for every overridden verdict, and propagating one correction through object, relation, place, scene, and mission layers at once.

The coverage paradox. Better scans can make density-based decomposition worse: at 99.6% coverage, gaps between adjacent objects close and DBSCAN chains 41% of the scene into one mega-cluster. Our measurements show this artifact accounts for only $\sim 12\%$ of the epoch’s absent nodes (5/43 recoverable by tighter re-splitting; 4/43 by Point-SAM refinement)—the rest is genuine change—but the effect argues for coverage-adaptive clustering or learned instance priors as the backbone’s next iteration, applied under the same flagged-candidate discipline as our Point-SAM rule.

Scope. Results come from one scanner, one building, and object-level change; structural change (walls, large fixtures) and cross-site generalization are untested. The three-epoch study mitigates but does not remove the two-scene limitation. The place layer’s regions are parameterized SLIC superpixels over the grid map: spatial cells rather than architecturally meaningful rooms, with the superpixel count fixed by hand; inferring wall- and doorway-aligned region boundaries is future work.

8 Conclusion

We presented TLS-SMF, a framework that turns registered terrestrial laser scans into a queryable, confidence-aware, and maintainable semantic knowledge graph for indoor robots. A deterministic geometric backbone makes decomposition reproducible and diffs meaningful; multi-view VLM verification with automatic escalation attaches calibrated trust to every label while retaining what it cannot resolve as explicitly unverified; and mint-once identities with two-pass matching turn re-scans into node-level upserts whose cost scales with actual change. Across two industrial scenes and a real three-epoch study of one room, the framework improved verification accuracy by up to 11 points through escalation, blocked hallucination traps by gating the data rather than instructing the model, preserved object identity across three scan epochs including a controlled physical change, and re-verified repeat scans at zero marginal VLM cost. Extending the loop beyond one-shot verification, automatic re-split gates and a structure-condition ensemble raised no-owner-input recall against an owner-audited 46-object inventory from 28% to 85% — including a nine-of-ten decomposition of a monitor wall the baseline had verified as a single cabinet — while the remaining type errors localized the owner’s role precisely: verifier bias is correlated across models, conditions, and view sets, so label truth enters through revisioned owner evidence over a single semantic store that also feeds the console, the mission mediator, and the simulation twin. The verifier-tier comparison makes the same point economically: a weaker model escalates more, resolves less, and costs more end-to-end.

Future work follows the limitations surfaced by our own evaluation: extending the remnant filter to non-vertical structure artifacts, coverage-adaptive decomposition to resolve the mega-cluster paradox, close-range or re-rendered imagery for small low-reflectance objects, structural-change updating beyond the object level, automatic place segmentation to close the loop with the DK-SMF lineage, and embodied validation in which a robot plans against the graph—including its unverified residue—in the loop.

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real multi-epoch update of the same room, latest transition (rev 2 $\rightarrow$ 3)
(cross-scan registration 4.2–5.1 cm RMSE; identity carried for specific-type objects)



Figure 15: Real multi-epoch update, latest transition (T2 $\rightarrow$ T3), top-down view. Colored points are the T3 segmented clusters inside the test-room boundary; yaw-rotated node footprints are colored by diff state (matched, moved with ID preserved, inserted, absent), with names on specific-type objects. Oversized footprints of merged mega-clusters (e.g., **machine_012**, Sec. 5.7) are faded.

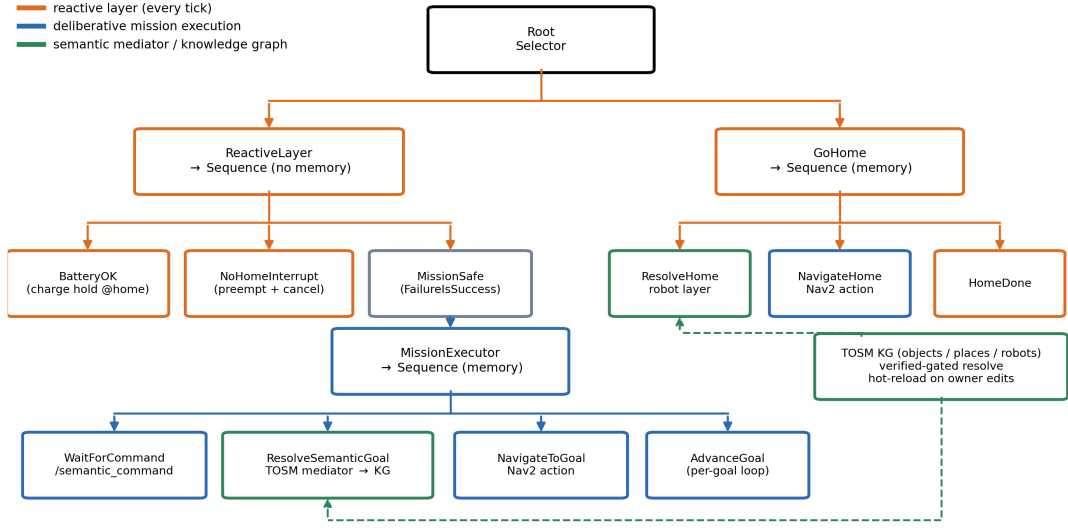


Figure 16: Hybrid deliberative/reactive mission behavior tree. The reactive layer re-evaluates battery and interrupt conditions every tick and can preempt navigation (canceling the active Nav2 goal); the deliberative branch resolves semantic commands through the TOSM mediator, which hot-reloads the knowledge graph when the management console edits it.

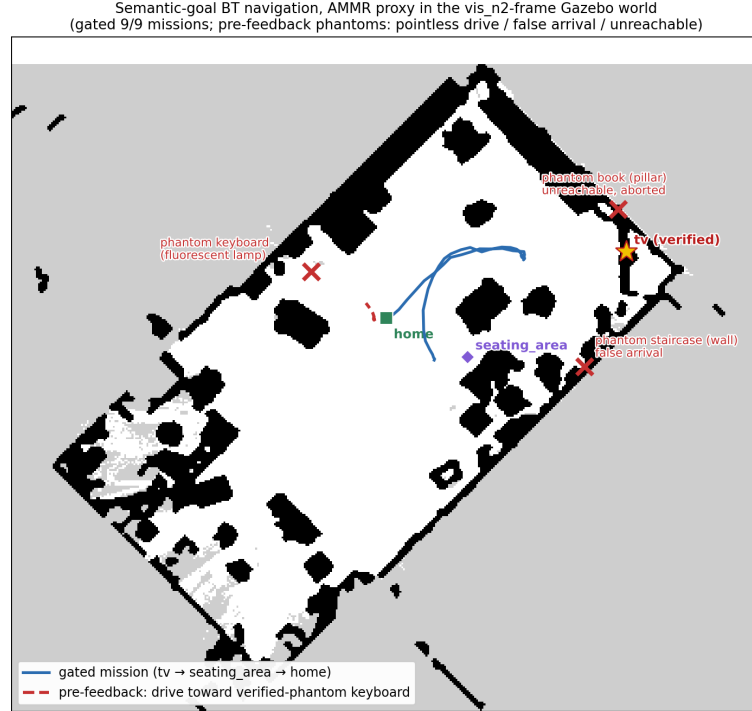


Figure 17: Semantic-goal navigation of the AMMR proxy in the Gazebo twin of the test room (world, map, and knowledge graph share the exploration-scan frame). Blue: a gated three-mission run (TV → seating area → home). Red dashed: against the pre-feedback graph the robot drives toward a verified-phantom keyboard — a fluorescent lamp; crosses mark the three phantom goals with their failure modes (pointless drive, false arrival, unreachable). All three are refused once structure filtering and owner feedback are applied.